

What is Blockchain?

Explorer Path · Blockchain Fundamentals

Blocks · Hashes · Distributed Records · When Blockchain Makes Sense



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From Records to Blocks

A blockchain organizes digital records into blocks and links them using cryptographic hashes, creating a sequence in which changes to earlier data can be detected.

1 Records

Digital records contain information about events, transactions, or other data.



Structured Records

2 Blocks

A blockchain groups records into blocks. Each block can contain data together with information used to connect it to the chain.



Grouped into Blocks

3 Blockchain

Blocks are linked in sequence using cryptographic hashes, creating a history in which changes to earlier data can be detected.



Cryptographically Linked



Key Takeaway:

A blockchain organizes records into cryptographically linked blocks, making changes to the recorded history detectable.

The Role of Hashes

A cryptographic hash function transforms input data into a fixed-size output called a hash. Hashes help link blocks and make changes to previously recorded data detectable.

1 What is a hash?

A cryptographic hash function transforms input data into a fixed-size output called a hash.



a3f5d2e1c9...
(256 bits)

2 Same input → Same hash

The same input produces the same hash when processed by the same hash function.



a3f5d2e1c9...
(256 bits)



a3f5d2e1c9...
(256 bits)

=

3 Small change → Different hash

Even a small change to the input will typically produce a very different hash.



a3f5d2e1c9...
(256 bits)

≠



7e9b4c8d1f...
(256 bits)

4 Why it matters in blockchain

Hashes are used to link blocks together. If the data in a block changes, its hash changes, which breaks the link to the next block and makes the change detectable.



Hash:
a3f5d2...

Hash:
7e9b4c...

Hash:
9c1d6e...

5 Important note on collisions

Hashes are designed to make collisions extremely difficult to find — but they do not mathematically guarantee that every possible input has a unique hash.



In practice, collisions are extremely rare (e.g., with SHA-256), but they are theoretically possible.



Key Takeaway:

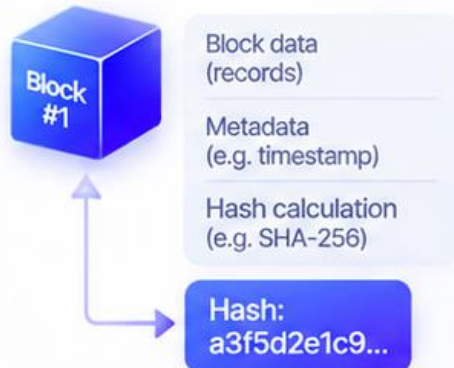
A hash acts like a compact cryptographic fingerprint of data, helping a blockchain detect changes without claiming that collisions are impossible.

Linking the Blocks

Blocks are linked in sequence using cryptographic hashes.
Each block contains its own data and a reference to the hash of the previous block.

1 Each block has its own hash

A block's hash is calculated from its contents (e.g., transactions, timestamp, other metadata).



2 A block references the previous block

Each block stores the hash of the previous block, creating a cryptographic link between them.



3 The links form a chain

Repeating this structure connects blocks in sequence and allows the recorded history to be checked.



4 What if an earlier block changes?

Changing the data in a block changes its hash. The hash stored in the next block no longer matches, revealing that the chain has been altered.



Key Takeaway:

By storing references to previous block hashes, a blockchain creates a linked history in which changes to earlier data can be detected.

Tamper-Evident History

Tamper-Evident Does Not Mean Immutable

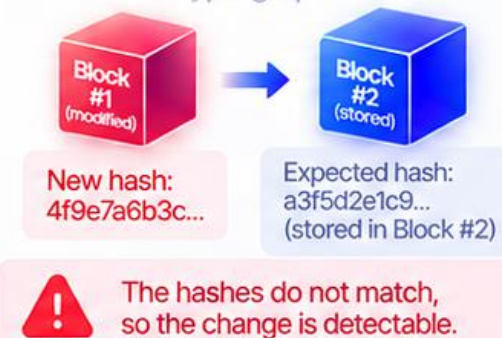
Blockchain data can be changed, but such changes become detectable.

Whether the change is accepted depends on the network's protocol and security model.

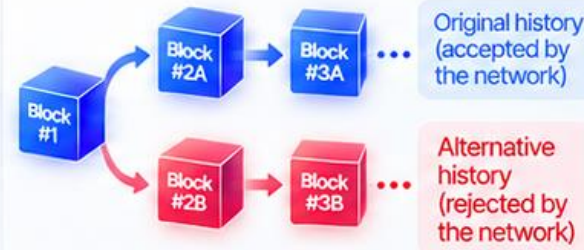
1 A change can be attempted
Blockchain data is not protected by a magical "cannot be changed" property. It is possible to modify data in a block (malicious participant).



2 The change becomes detectable
Changing the data in a block changes its hash. The hash stored in the next block no longer matches, breaking the cryptographic link.



3 Hashing alone is not enough
Hashes reveal inconsistencies, but they do not decide which version of the ledger is correct. Multiple versions of the data could exist, especially if some participants try to introduce an alternative history.



4 The network protocol matters
Consensus rules and the network's security model determine how participants agree on valid history and respond to competing versions of the ledger.



Key Takeaway:

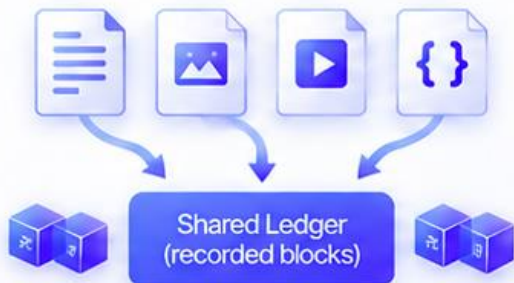
Hash-linked blocks make tampering evident.

Resistance to rewriting blockchain history depends on more than hashing alone.

One Ledger, Multiple Participants

Blockchain allows multiple participants to maintain and verify a shared record without relying on a single database as the only source of truth.

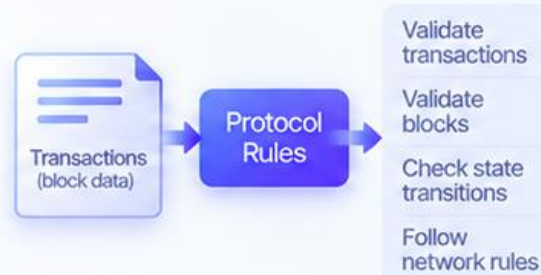
- 1 Shared Record**
Multiple participants can access or verify a common history of recorded transactions or events.



- 2 Replicated Across the Network**
Depending on their role and the network design, participating nodes may store and validate some or all of the ledger data.



- 3 Protocol Rules**
Participants follow common protocol rules to validate transactions, blocks, and the state of the ledger.



- 4 Agreement**
Consensus mechanisms help the network determine which valid history participants should accept.



Important note:

Blockchain networks differ: who can participate, what data each node stores, and how agreement is reached depend on the network architecture.



Key Takeaway:

A blockchain can provide a shared, verifiable record across multiple participants — but its degree of decentralization depends on how the network is designed and governed.

What Blockchain Gives Us

By combining cryptographic hashing, linked blocks, and a distributed network of participants, blockchain enables several important properties.

1 Tamper Evidence

Cryptographic links help make changes to previously recorded data detectable.



2 Traceability & Auditability

Recorded transactions and events can provide a history that authorized participants can inspect and verify, depending on the network design.



Block #125
Block #124
Block #123
...

3 Shared State

Multiple participants can work with a common view of recorded data according to the network's protocol rules.



4 Reduced Reliance on a Single Authority

In appropriately designed networks, control and verification can be distributed across multiple participants rather than depend on one central operator.



Important note:

These are potential properties, not automatic guarantees.
Their strength depends on the blockchain architecture, governance, and security model.



Key Takeaway:

Blockchain can provide shared verification, traceability, and tamper evidence — when the network is designed to support those properties.

What Blockchain Does NOT Guarantee

Blockchain can provide important properties, but it does not automatically solve all problems. There are important limitations to understand.

1 Correct Data

Blockchain can protect the integrity of recorded data, but it cannot determine whether the data was true or accurate when it was submitted.



Integrity ≠ Truth

The system records what is submitted, not whether it is correct.

2 Automatic Trust

Cryptography can help verify records and actions, but trust still depends on participants, governance, software, and the information entering the system.



Trust depends on people, processes and governance, not just technology.

3 Complete Privacy

Blockchain does not automatically provide privacy. Depending on the network design, recorded data or transaction activity may be visible to many participants or even publicly accessible.



Visibility depends on the network type and design (e.g., public, permissioned, private data).

4 Perfect Security

Blockchain does not eliminate vulnerabilities in applications, smart contracts, keys, endpoints, or operational processes.



Security still requires careful design, development, and operational practices.

5 Permanent Correctness

Incorrect information may become part of the recorded history. Corrections can be added, but the treatment of earlier records depends on the system design.



Corrections can be recorded, but earlier data may remain in the history, depending on the network's rules and design.



Key Takeaway:

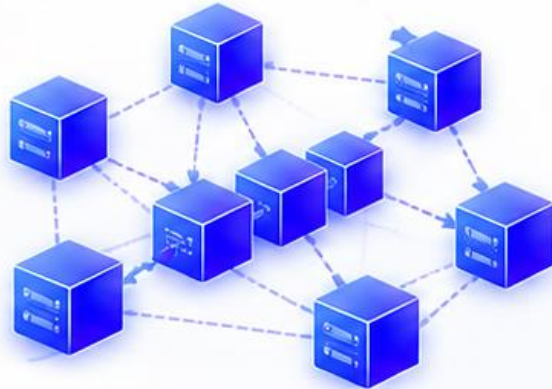
Blockchain can strengthen integrity and verifiability, but it does not make data true, systems perfectly secure, or trust unnecessary.

Blockchain Is Not Cryptocurrency

Blockchain is a general technology for recording and verifying data. Cryptocurrency is one important application of that technology, but blockchain can support many other use cases.

1 Blockchain

A blockchain is a technology for recording and verifying data across a network of participants according to shared protocol rules.



What can it support?

 Digital assets and cryptocurrencies Supply-chain records Credentials and certificates

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 Identity-related proofs Smart-contract applications

... and more use cases



Cryptocurrency is one important application of blockchain technology — but blockchain has **many other possible uses.**

2 Cryptocurrency

A cryptocurrency is a digital asset designed for transferring or representing value using a distributed ledger or related decentralized infrastructure.



Examples

 Bitcoin (BTC) Ether (ETH)

and many other cryptocurrencies



Key Takeaway:

Blockchain is the underlying technology; cryptocurrency is one category of applications built using blockchain and related distributed-ledger technologies.

When Does Blockchain Make Sense?

Blockchain is useful when multiple parties need to share and maintain a trusted record without relying on a single central authority.

It is not the right solution for every problem. Understanding the right use cases helps us apply it effectively.

✓ Blockchain is a good fit when...



Multiple Participants

Several independent parties need to access and update the same data.



Need for a Trusted Record

The data should be tamper-evident and auditable by all participants.



Low Trust Environment

Participants may not fully trust each other and there is no single central authority.



Long-Term Transparency

The data needs to be available and verifiable over a long period of time.

✗ Blockchain is usually NOT needed when...



Single Organisation

A traditional database is simpler and more efficient.



High-Speed Transactions

Real-time, high-volume applications may be better served by traditional systems.



Sensitive Private Data

Storing sensitive data directly on a public blockchain is not appropriate.



No Need for Shared Control

If a central authority is acceptable and trusted, blockchain may add unnecessary complexity.



Key Takeaway:

Use blockchain when you need a shared, tamper-evident and auditable record across multiple parties with limited mutual trust. Do not use it by default — simpler solutions may be more appropriate.

From Concept to Exploration

You now understand the basic structure of a blockchain. The next step is to observe it, experiment with it, and see how changes affect the chain.

1

Observe

Anders Brownworth Blockchain Demo

Explore an interactive demonstration of blocks, hashes, and blockchain structure.

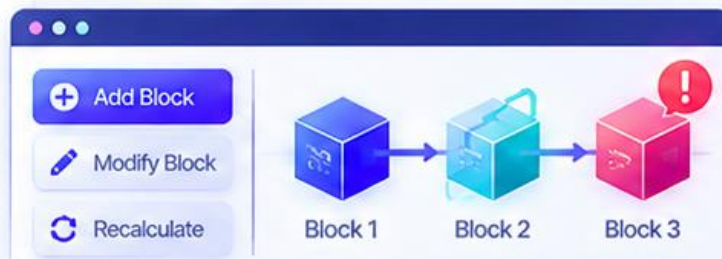


2

Experiment

Web3Edu Blockchain Simulator

Build a simple chain, modify a block, and observe how hash relationships change.



3

Reflect

Connect what you observe with the concepts from this lesson: hashes, linked blocks, tamper evidence, and shared records.

- ☒ Hashes
- ☒ Linked blocks
- ☒ Tamper evidence
- ☒ Shared records



Suggested sequence:



Learn

Review the key concepts from this lesson



Observe

Explore the interactive demo



Experiment

Try the simulator and make changes



Assess

Check your understanding with the lesson activities



Key Takeaway:

Understanding becomes stronger when you move from concepts to hands-on exploration.

Key Takeaways

Blockchain is a powerful technology with unique properties, but it is not a solution for every problem.

Understanding both its capabilities and its limitations helps us use it correctly and set realistic expectations.

1 Shared, Tamper-Evident Record

Blockchain records data in blocks linked with cryptography, making unauthorised changes difficult.



2 Transparency and Auditability

Participants can verify the history of transactions, increasing trust and accountability (subject to the network design).



3 Not the Same as Cryptocurrency

Blockchain is the underlying technology. Cryptocurrency is one possible use case, not the same thing.



4 Benefits with Trade-offs

Blockchain can improve trust, reduce reliance on a single authority and enable new applications, but it also has trade-offs (e.g., performance, cost, complexity, privacy).



5 Use it When it Fits

Blockchain makes sense when multiple parties need a shared, trusted record. For many other problems, simpler solutions (e.g., traditional databases) are more appropriate.



6 Explore and Practice

The best way to build understanding is to experiment. Use the interactive demo and the Web3Edu simulator to see how blockchain works in practice.



Final Message:

Blockchain is a valuable tool when its strengths match the requirements of the problem. Use it with a clear understanding of its limitations and always consider simpler alternatives.

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You've Completed the Visual Lesson

You now have the foundations to explain what a blockchain is, how blocks and hashes are linked, and what blockchain can — and cannot — provide.



Next in LM01



1. Explore the demo

See an interactive demonstration of blocks, hashes, and blockchain structure.



2. Experiment with the simulator

Build a simple chain, modify a block, and observe how hash relationships change.



3. Complete the assessment

Check your understanding with the lesson activities and questions.



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Key Takeaway:

This visual lesson is your starting point. Now explore, experiment, and keep learning on Web3Edu.